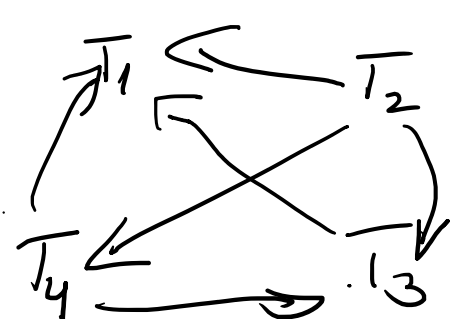


	T1	T2	T3	T4
		R(A) W(A)		
				R(A) W(A) W(E) R(E)
			R(E) W(E)	
		R(B) R(C)		
			R(A) W(A)	
R(A)		W(B)		



The graph is acyclic, is conflict serializable.

Let S be this schedule. Compute its conflict relation. Is S conflict serializable? Justify.
Find a serial schedule equivalent to S

LSN	prevLSN	transId	type	pageId
1	*	T10	update	P1
2	*	T12	update	P1
3	1	T10	update	P1
4	3	T10	update	P2
5	4	T10	update	P3
6	*	T11	update	P1
7	2	T12	update	P2
8	6	T11	update	P3

Write Transaction Table and Dirty Page Table for the table above.

Trans Table

TransId	Status	LastLSN
T10	in progress	5
T11	in progress	8
T12	in progress	7

Dirty table

Page Id	RecLSN
P1	1
P2	4
P3	5

Page Id	RecLSN
P ₁	1
P ₂	4
P ₃	5

Let R_1, R_2 be two relations. $R_1 \rightarrow 200,000$ records, a page holds 50 R_1 records. $R_2 \rightarrow 10,000$ records, a page holds 200 R_2 records.

a) 102 buffer pages available. R_1, R_2 not sorted

• Compute the cost of $R_2 \bowtie_{R_2.J \bowtie R_1.J} R_1$ using block nested loops join and sort-merge join; R_2 is the outer relation (the dominant part in the family). Assume each partition is scanned once during merging phase of sort-merge join.

$R_1 \rightarrow 200,000$ records
50 pe

4000 pages $B = 102$

$R_2 \rightarrow 10,000$ (outer)

200 pe

50 pages

Block nested loops join

$$\text{no. of blocks} = \left\lceil \frac{50}{100} \right\rceil = 1$$

$$\text{Total cost} = 50 + 1 * 4000 = 4050$$

Sort merge join $\rightarrow$

$$R_1 \text{ sort cost: } 2 * 4000 * \left(\left\lceil \log_{101} \left\lceil \frac{4000}{100} \right\rceil \right\rceil + 1 \right) = 8000 * 2 = 16000$$

$$R_2 \text{ sort cost: } 2 * 50 \left(\left\lceil \log_{101} \left\lceil \frac{50}{100} \right\rceil \right\rceil + 1 \right) = 100 * 1 = 100$$

$$\text{Cost} = 16000 + 4000 + 50 = 20050 \text{ i/o}$$

cost of sort-merge join.

• What is the minimum number of pages in Buffer Pool that keeps the cost unchanged for both blocked nested loops join and sort merge join? Justify.

$$\log_{101} \left\{ \begin{aligned} \left\lceil \frac{4000}{100} \right\rceil &= 40 & \left\lceil \frac{4000}{50} \right\rceil &= 80 \\ \left\lceil \frac{50}{100} \right\rceil &= \frac{1}{2} \rightarrow 1 & \frac{50}{50} &= 1 \end{aligned} \right.$$

$B \geq 1$

The minimum number of Buffer pages would be 50 as it won't change the cost of the joins.

c) R1 is stored at Resita, R2 at Galati. Compute the cost of $R2 \bowtie_{R2,30} R1$ using page oriented nested loops join in Galati without caching. R2 is the outer relation, the query site is Baia Mare and the result $R2 \bowtie_{R2,30} R1$ has 5000 pages.

$R1 = 4000$ Resita
 $R2 = 50$ Galati

Cost of join query at Galati would be cost of nested loops join + shipping of pages from Resita to Galati.

oriented nested loops join:

$$\text{Cost} = 4000 + (50 + 50 + 4000) = 204000$$

Final cost (+ shipping from Galati to Baia Mare)

$$= 204000 + 5000 = \underline{209000}$$

$$= 204\,000 + 5000 \approx 210\,000$$

III 1 Under READ COMMITTED layer:

- a) You can't acquire X locks
- ☒ b) You can acquire X locks
- ☒ c) You can acquire S locks
- d) You can't acquire S locks
- e) none

2 In ARIES, the redo phase:

- a) starts at the most recent checkpoint
- ☒ b) starts at the smallest recLSN in Dirty Page Table
- c) determines the starting point for undo phase
- d) determines the starting point for analysis phase
- e) none

3 The reduction factor for ~~the~~ condition $Age > 0$ assuming data is uniformly distributed and there is an index J on Age , can be estimated by:

- a) $(20 - J_{High}(J)) / (J_{Low}(J) - J_{High}(J))$
- b) $(J_{High}(J) - 20) / (J_{Low}(J) - J_{High}(J))$
- c) $(20 - J_{High}(J)) / (J_{High}(J) - J_{Low}(J))$
- ☒ d) ~~$(20 - J_{High}(J))$~~ $(J_{High}(J) - 20) / (J_{High}(J) - J_{Low}(J))$
- e) none

4] Let R be a relation and $c_1, \dots, c_m$ selection conditions
 σ - sigma (selection operator)

a) $\sigma_{c_1}(\sigma_{c_2}(\sigma_{c_3}(R))) \equiv \sigma_{c_1 \wedge c_2 \wedge c_3}(R) \equiv \sigma_{c_1}(\sigma_{c_3}(\sigma_{c_2}(R)))$

b) $\sigma_{c_1 \wedge c_2 \wedge c_3}(R) \equiv \sigma_{c_1}(\sigma_{c_2}(\sigma_{c_3}(R)))$

c) $\sigma_{c_1 \vee c_2 \vee c_3} \equiv \sigma_{c_1}(\sigma_{c_2}(\sigma_{c_3}(R)))$

d) $\sigma_{c_1 \wedge c_2 \wedge c_3}(R) \equiv \sigma_{c_2}(\sigma_{c_3}(\sigma_{c_1}(R)))$

e) none

5] choose access paths that can be used for a query with ~~an~~ relation in the FROM clause:

a) single index access path

b) multiple index access path

c) sorted index access path

d) index only access path

6] The transactions T_1, T_2 execute serially on a consistent database state.

a) The database is in a consistent state after execution

b) The database is in an inconsistent state after execution

c) T_2 cannot write operations

d) T_2 cannot read operations

e) none

7] "A log fragment was given". T_2 update described by log with LSN 5 is undone. The undoNextLSN field of the corresponding CLR has a value of

a) 5 b) 4 c) 3 d) 2 e) none

8] RSA uses a) a public key and a secret key

b) only a public key c) only a secret key

d) $\log_2 1000$ secret keys e) none

9) Concurrent transactions T_1, T_2 can lock same object in the following manner:

a) T_1 has X lock, T_2 S lock

b) $-|| - X \text{ lock} - || - X \text{ lock}$

c) S lock S lock

d) no main stu

e) none

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algorithm

11) Relation R is sorted with external merge sort. R has N pages. There are B buffer pages. The first phase of the algorithm produces:

a) $\left\lceil \frac{B}{N} \right\rceil$ sorted runs of N pages each

b) $\left\lceil \frac{N}{B} \right\rceil$ sorted runs of B pages each

c) $\left\lceil \frac{N}{N} \right\rceil$ sorted runs of B^3 pages each

d) $\left\lceil \frac{B}{N-1} \right\rceil$ sorted runs of N^3 pages each

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- 12] Join implementation techniques:
- a) block-nested loops join based on iteration techniques
 - b) block-nested loops join based on partitioning technique
 - c) hash join based on iteration
 - d) hash join based on partitioning
 - e) none

- 13] ACID stands for:
- a) ?
 - b) acid..
 - c) atomicity, consistency, isolation, durability
 - d) ?
 - e) none

- 14] The following hold:
- a) Dirty reads can occur under READ UNCOMMITTED
 - b) Dirty reads cannot occur under REPEATABLE
 - c) Unrepeatable reads cannot occur under SERIALIZABLE
 - d) Unrepeatable reads can occur under READ COMMITTED
 - e) none